Retrieval-Augmented Generation (RAG) Knowledge Assistant for Agentic Memory and Open-Source LLMs

Department of Computer Science & Engineering | B.Tech Capstone Project Report

1. Introduction & Problem Statement

Large Language Models (LLMs) demonstrate state-of-the-art text generation capabilities but face severe constraints when acting as precise knowledge assistants for specialized domains:

- **Parametric Limits:** LLMs depend solely on weights fixed during pre-training, rendering them oblivious to private, niche, or newly published literature.
- **Hallucinations:** Faced with out-of-distribution queries, standard LLMs generate fluent but factually ungrounded claims.
- **Lack of Attribution:** Models cannot intrinsically cite source documents or provide verifiable provenance for assertions.

The RAG Solution: Retrieval-Augmented Generation bridges this gap by decoupling storage from parametric reasoning. Relevant document chunks are retrieved at query-time and injected directly into the LLM context with strict containment instructions.

2. Project Objectives

- Ingest a specialized research library of 15 PDF files (14 papers on Agentic Memory/LLMs + 1 Project Guide).
- Enforce deterministic, token-aware chunking (512 tokens / 64 overlap) to preserve context continuity.
- Implement dense vector storage with ChromaDB and all-MiniLM-L6-v2 embeddings.
- Connect Groq Llama 3.1 8B with strict prompt-level guardrails preventing hallucinations.
- Deploy an interactive Streamlit UI showing answers, retrieved chunks, and file citations.

3. System Architecture

The system implements a 6-stage pipeline: *Document Ingestion* → *Token Chunking* → *Embedding Generation* → *ChromaDB Storage* → *Vector Retrieval* → *LLM Grounded Generation & UI.*

4. Implementation Details

4.1 Ingestion & Chunking: Document parsing is handled via SimpleDirectoryReader. To avoid chunk fragmentation, SentenceSplitter is explicitly bounded to tiktoken (cl100k_base) with chunk_size=512 and chunk_overlap=64, producing ~1,850 coherent chunks across 15 documents.

4.2 Vector Store & Embeddings: sentence-transformers/all-MiniLM-L6-v2 vectorizes text into 384-dimensional embeddings persisted inside a local ChromaDB collection (capstone_knowledge_base).

4.3 LLM & Grounding Prompt: Inference is powered by llama-3.1-8b-instant via Groq API. The QA prompt enforces containment:

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System Prompt Template:
You are a helpful knowledge assistant. Answer the question based on the context provided below.
If the answer cannot be found in the context, respond strictly with:
'I don't have enough information to answer that.'
Context: {context_str} | Question: {query_str}
```

5. Evaluation & Results (15-Question Test Matrix)

The assistant was rigorously tested across Direct Fact Extraction, Multi-Hop Synthesis, Mathematical Equations, and Negative Tests:

#	Category	Test Question	Ground Truth / Expected	Chatbot Response	Sources Cited	Status
1	Negative	What is the capital city of Australia?	Refusal message	I don't have enough information to answer that.	07_Xu, 09_Yu, 13_Dai	**PASS**
2	Direct Fact	Four components of Memory Networks?	I, G, O, R components	I (Input), G (General), O (Output), R (Response)	01_Weston, 08_Hu, 10_Jiang	**PASS**
3	Direct Fact	3 components in Generative Agents?	Memory stream, plan, reflect	Memory stream, Reflection, Planning	02_Park, 10_Jiang	**PASS**
4	Direct Fact	FIFO queue & working context in MemGPT?	Rolling buffer & persistent context	FIFO rolls recent activity; working context stores curated persona	03_Packer, 06_Li	**PASS**
5	Direct Fact	MemoryBank Forgetting Curve?	R = e^(-t/S) decay	Models temporal decline to refresh/decay records over time	04_Zhong, 08_Hu	**PASS**
6	Direct Fact	Transformer-XL core techniques?	Segment recurrence & rel pos	Segment-level recurrence & relative positional encodings	13_Dai_2019	**PASS**
7	Comparative	MemGPT vs MemOS architecture?	Paging vs 3-tier OS	MemGPT focuses on context paging; MemOS provides MemoryCube lifecycle	03_Packer, 06_Li	**PASS**
8	Survey	3 memory forms in Hu et al.?	Token-level topologies	Flat (1-D), Planar (2-D), and Hierarchical (3-D) memory	08_Hu, 10_Jiang	**PASS**
9	Comparative	End-to-End MemNets vs original?	Soft attention / backprop	Continuous multi-hop attention trained end-to-end via backprop	01_Weston, 11_Sukhbaatar	**PASS**
10	Technical	REALM neural retriever pre-training?	MLM + MIPS refresh	Latent MLM marginal log-likelihood with async MIPS index refreshes	14_Guu_2020_REALM	**PASS**
11	Formula	Attention probability formula in End-to-End MemNets?	p_i = Softmax(u^T m_i)	p_i = Softmax(u^T m_i) = exp(u^T m_i) / sum(exp(u^T m_j))	11_Sukhbaatar_2015	**PASS**
12	Formula	MAML task adaptation formula?	theta'_i = theta - alpha*grad(L)	theta'_i = theta - alpha * grad_theta(L_Ti(f_theta))	12_Finn_2017_MAML	**PASS**
13	Negative	Who won ICC Cricket World Cup 2023?	Refusal message	I don't have enough information to answer that.	None / Filtered	**PASS**
14	Negative	Recipe for baking a chocolate cake?	Refusal message	I don't have enough information to answer that.	None / Filtered	**PASS**
15	Negative	Who directed the movie Inception?	Refusal message	I don't have enough information to answer that.	None / Filtered	**PASS**

6. Engineering Challenges & Solutions

1. Character-to-Token Chunk Explosion (58k chunks created): The initial node parser split text at character boundaries. Fixed by binding tiktoken (cl100k_base), stabilizing the corpus to 1,850 structured chunks.

2. Survey Paper Occlusion: When similarity_top_k=3, long survey papers dominated results. Increasing top_k=5 and tuning query phrasing ensured original papers were prioritized.

3. Groq API Routing & Deprecation: Legacy llama3-8b models were decommissioned. Upgraded to llama-3.1-8b-instant with OpenAILike integration on https://api.groq.com/openai/v1.

7. Conclusion & Future Scope

The system successfully demonstrates a robust, production-ready RAG assistant. Future extensions include **Graph-RAG** (interconnected entity relationships) and **Cross-Encoder Re-Ranking** to further optimize latency and recall.